

CMSC 104 Section 2

Spring 2026

Sample Quiz 2

Answer Key

Instructions:

This quiz is to be completed on paper, using a pen or pencil. Please make sure you write your name on the paper. When you are finished you may turn in your paper and leave the room.

This quiz is “open-book, open-notes.” You MAY use any of the following: your laptop or phone, a book; your notes; the instructor’s notes and code from the GitHub site; an IDE such as IDLE or PyCharm (if you wish to test your code to make sure your answer is correct). You MAY NOT use: any other person, whether in this classroom or not (e.g., you may not text your friend the computer science major), or any AI-support source such as ChatGPT, Gemini, CoPilot, Claude Code, etc. If you have a question about this, please raise your hand and ask.

Section 1: True/False and Multiple Choice (40 points)

There are 10 questions in this section. Each is worth 4 points. No partial credit is given for any question in this section. If you change your answer during the quiz, please make sure you clearly indicate which is your final answer.

1. Which of the following is NOT a reason for using functions in a Python program?
 - a. Functions allow you to reuse code by importing your own Python programs into other programs
 - b. Functions make it easier write and test code, a module at a time
 - c. **Functions make a program much more complex because of the needless function definitions and function calls**
 - d. All of the above are good reasons not to use functions in a Python program

2. Which of the following is not a valid Python variable name?
 - a. tooCoolforSchool
 - b. ToO_CoOl_4_Skool
 - c. Too_c00l_4_sk00l
 - d. **All of these are valid Python variable names**

3. True or False: in Python it is permissible to write an if-else statement with no code under the "if" branch. That is, the following would be legal:

```
if age > 21:
else:
    print("sorry, you are still a minor")
```

- a. True
 - b. False**
4. True or False: "true" is a legal value for a Boolean variable in Python
- a. True
 - b. False**
5. What happens if you have a Python function call with 2 arguments, but the function definition has 3 parameters?
- a. Python puts the values from the two arguments into the first two parameters; the last parameter gets the value 0
 - b. The program crashes because the number of arguments and the number of parameters must be the same**
 - c. Python puts the values from the two arguments into the last two parameters; the first parameter gets the value None
 - d. None of the above answers is correct
6. What does the following Python code produce as output?
- ```
product = 1
for i in range(1,4,1):
 product = product * i

print(product)
```
- a. 24
  - b. 6**
  - c. 0
  - d. None of the above answers is correct
7. In general, what is the best way to develop computer software?
- a. Write a little; test a little**
  - b. Write the whole program at once and hope like heck it works
  - c. Just steal it from somebody else and hope they don't sue you for theft
  - d. Seriously? Just choose a.

8. True or False: you can have an if-elif-else statement without an else. That is, the statement

```
if age > 21:
 print("You have reached the age of majority")
elif age > 18:
 print("You can vote but you are not fully an adult")
```

is legal in Python

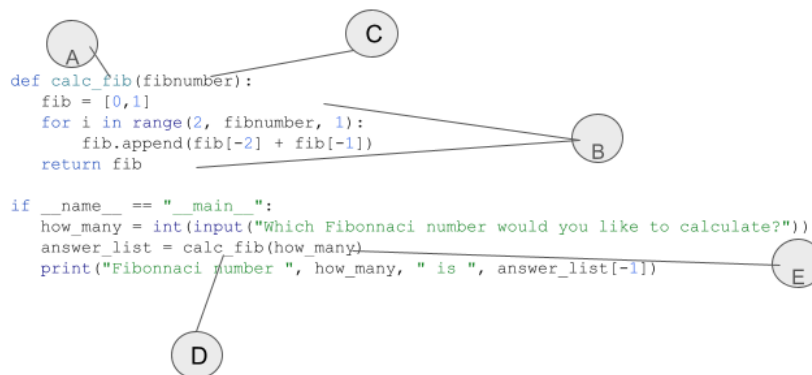
- a. **True**
  - b. False
9. What is the value printed by the following line of code?
- ```
print ((5//4)**(3 + 2%4))
```
- a. **1**
 - b. 5
 - c. 12
 - d. Nothing is printed; the statement is an error
10. True or False: when you import one of your own files into a Python program, you can use any function from the imported program in the new one, but you can't use code that was in the main program of the imported file.
- a. **True**
 - b. False

Section 2: Short answer (40 points)

There are 8 questions in this section. Each is worth 5 points. Partial credit IS given for questions in this section, so please make an attempt to answer each question.

Note: “short answer” means one or two sentences. It should not take you more than that to answer a question. Continuing to write paragraph after paragraph is probably not going to help make your answer more correct.

11. Identify the parts of a function definition and call



Argument(s)
Parameter(s)
Function call
Function body
Function name

Arguments - E; Parameters- C; Function Call - D; Function Body - B; Function Name - A.

12. Suppose that your program reaches a point where, if a boolean variable `Error_code` is equal to `True`, you want to print an error message, but if it's `False` you don't want to take any action. Would you use an `if`, an `if-else`, or an `if-elif-else`?

If. That is designed to be used for just this situation.

13. Write a “for i” loop that sums the first 20 integers - the numbers from 1 to 20.

```
sum = 0  
for i in range(1,21,1):  
    sum = sum + i
```

14. What is a Boolean variable?

A variable that can have only two possible values: True and False

15. Is your computer's keyboard an example of software or hardware?

Hardware

16. What will the following Python code output?

```
sum = 0
for i in range(10,0,-1):
    sum = sum + i
print (sum)
```

55

17. Write an if-elif-else statement that does the following: suppose that x and y are both Boolean variables. If both x and y are True, your statement prints out "this is an AND statement." If both x and y are False, your statement prints out "this is a not AND statement." In any other case your statement prints out "you need an exclusive-or"

```
if x and y:
    print("this is an AND statement")
elif not x and not y:
    print("this is a not AND statement")
else:
    print("you need an exclusive or")
```

18. Write two Python statements that import the library called "math" and then use the square root function, sqrt, to assign the square root of 26 to the variable x.

```
import math
x = math.sqrt(26)
```

Section 3: Programming (20 points)

There are two questions in this section; each is worth 10 points. Partial credit IS given for questions in this section.

You do not need to write comments with your code, unless you think that it will help the grader more clearly understand exactly what you're trying to accomplish.

19. Write a program to assign a letter grade based on a user's numeric test score. The test score is an integer between 0 and 20 inclusive. Your program should:

- Ask user for test score, reminding the user that the score ranges from 0 to 20.
- if the Score ≥ 15 assign an "A"
- if the score is from 10 to 14 assign a "B"
- any other score, assign a "C".

```
score = int(input("Please enter the user's test score. Valid scores  
are from 0 to 20"))  
if score >= 15:  
    grade = "A"  
elif score >= 10 and score <= 14:  
    grade = "B"  
else:  
    grade = "C"
```

Alternatively:

```
elif 10 <= score <= 14:  
    Grade = "B"
```

20. Write a function to calculate the area of a trapezoid. A trapezoid is a four-sided figure that has two sides parallel to each other but are of different lengths; the other two sides are thus not parallel to each other. To calculate the area,

- add the lengths of the two parallel sides.

- take one-half of this value

- multiply this by the height - the distance between the two parallel sides

Your function should take three parameters: the length of one parallel side; the length of the other parallel side; and the height. Your function should return one value - the area.

You don't have to write the whole program; just this function.

```
def trap_area (p1_length, p2_length, height):  
    area = (p1_length + p2_length)/2 * height  
    return area
```